ENGINEERING GEOLOGY LABORATORY

(23A01401P) · II Yr B.Tech CE – II Sem · JNTUA R23 · Answers & Record Notes

0. Lab Examination Pattern – How to Score

1. Description & identification of **FOUR minerals**
2. Description & identification of **FOUR rocks** (igneous + sedimentary + metamorphic)
3. **ONE** question – interpret a geological map + draw geological section
4. **TWO** simple strike & dip problems
5. **Bore-hole** problem
6. **Project report** on geology (field work)

> Write for every specimen in this order : **Name → Group/Class → Colour → Streak → Lustre → Hardness → Cleavage/Fracture → Form/Texture → Special test → Uses / Engg. importance.**

Experiment 1 – Physical Properties of Minerals

Aim : To identify rock-forming and ore-forming minerals megascopically (hand specimen) from their physical properties.

Apparatus : Mineral specimens, streak plate, pen-knife / steel nail, copper coin, glass plate, dil. HCl, hand lens, magnet.

Physical properties – definitions

- **Colour** – appearance in reflected light (unreliable for many minerals); **Streak** – colour of powder on unglazed porcelain (reliable).
- **Lustre** – shine : metallic, sub-metallic, vitreous, resinous, pearly, silky, greasy, earthy.
- **Hardness** – resistance to scratching (Mohs' scale 1–10).
- **Cleavage** – breaking along smooth planes of weakness; **Fracture** – irregular breaking (conchoidal, uneven, fibrous).
- **Specific gravity** – density relative to water; **Form / habit** – crystal shape (prismatic, fibrous, massive…).
- **Special** – magnetism, effervescence with HCl, feel (soapy, greasy), taste, elasticity, tenacity.

H	Mineral	Test
1	Talc	Nail scratches easily, soapy
2	Gypsum	Finger nail scratches

H	Mineral	Test
3	Calcite	Copper coin scratches
4	Fluorite	Easily with knife
5	Apatite	Knife with difficulty
6	Orthoclase	Cannot be scratched by knife; scratches glass faintly
7	Quartz	Scratches glass & steel
8	Topaz	–
9	Corundum	–
10	Diamond	Hardest

Mnemonic : The Girls Can Fly Above Our Quiet Tall Class Doors.

1(a) Rock-Forming Minerals – Identification Answers

Mineral / Formula	Colour · Streak · Lustre	H · SG	Cleavage / Fracture / Form	Identifying features · Uses
Quartz SiO_2	Colourless, white, milky, rose, smoky · streak white · vitreous (greasy on fracture)	7 · 2.65	No cleavage; **conchoidal** fracture; hexagonal prism with pyramid	Scratches glass; transparent–translucent. Glass, abrasive, aggregate, ornaments (amethyst, citrine)
Orthoclase (Feldspar) $KAlSi_3O_8$	Pink, flesh, white · white · vitreous	6 · 2.56	**2 cleavages at 90°**; monoclinic; Carlsbad twin	Pink colour + right-angle cleavage. Ceramics, glass, porcelain. (Microcline : green "amazonite")
Plagioclase (Feldspar) $NaAlSi_3O_8$–$CaAl_2Si_2O_8$	White, grey, bluish · white · vitreous–pearly	6–6.5 · 2.62–2.76	2 cleavages at ≈ **86°**; fine parallel **striations** (twinning)	Striations on cleavage face. Rock-forming (basalt, gabbro); ceramic
Garnet $(Fe,Mg,Ca)_3Al_2(SiO_4)_3$	Deep red, brown, pink · white · vitreous / resinous	6.5–7.5 · 3.5–4.3	No cleavage; sub-conchoidal; **dodecahedral** crystals	Red rounded crystals in schist/gneiss. Abrasive (garnet paper), gem

Mineral / Formula	Colour · Streak · Lustre	H · SG	Cleavage / Fracture / Form	Identifying features · Uses
Muscovite (Mica) $KAl_2(AlSi_3O_{10})(OH)_2$	Colourless, silvery, pale · white · pearly / silky	2.5–3 · 2.8–3.0	**Perfect basal** cleavage; thin **elastic** sheets	Peels into transparent sheets. Electrical & heat insulator, paints
Biotite (Mica) $K(Mg,Fe)_3AlSi_3O_{10}(OH)_2$	Black, dark brown · white/grey · pearly–vitreous	2.5–3 · 2.7–3.1	Perfect basal cleavage; elastic flakes	Black shiny flakes in granite, gneiss, schist; weathers to vermiculite
Talc $Mg_3Si_4O_{10}(OH)_2$	White, apple-green, grey · white · pearly / greasy	1 · 2.7–2.8	One perfect cleavage; foliated / massive (soapstone)	**Soapy feel**, scratched by nail; flexible not elastic. Talcum powder, paper, ceramics
Chlorite $(Mg,Fe)_5Al(AlSi_3O_{10})(OH)_8$	Green (dark) · pale green · pearly / dull	2–2.5 · 2.6–3.3	One perfect cleavage; scaly flakes, **flexible but not elastic**	Green flakes in schist/greenstone; metamorphic indicator
Olivine $(Mg,Fe)_2SiO_4$	Olive-green, yellow-green · white · vitreous	6.5–7 · 3.3–3.4	Poor cleavage; conchoidal; granular	Green glassy grains in basalt / gabbro. Refractory, peridot gem
Kyanite Al_2SiO_5	Blue (also white, grey) · white · vitreous–pearly	**5 ∥ length; 7 ⊥ length** · 3.5–3.7	One perfect + one good cleavage; bladed crystals	Blue blades in schist/gneiss; **anisotropic hardness.** Refractories, porcelain
Asbestos (chrysotile) $Mg_3Si_2O_5(OH)_4$	White, grey, greenish · white · **silky**	2.5–4 · 2.4–2.6	**Fibrous**, flexible fibres, splits into threads	Heat-proof, acid-proof; asbestos sheets, brake lining. Carcinogenic dust
Tourmaline (Schorl) complex Na-Fe-Al borosilicate	Black (also green, pink) · white/grey · vitreous	7–7.5 · 3.0–3.2	No cleavage; conchoidal; **striated 3-sided prism**	Striated rounded-triangle prism in pegmatite/granite. Gem, pressure gauges

Mineral / Formula	Colour · Streak · Lustre	H · SG	Cleavage / Fracture / Form	Identifying features · Uses
Calcite $CaCO_3$	Colourless, white, grey · white · vitreous	3 · 2.71	**3 cleavages not at 90°** (rhombohedral)	**Effervesces with dil. HCl**; double refraction. Limestone, cement, lime
Gypsum $CaSO_4 \cdot 2H_2O$	White, colourless, pink · white · silky / pearly / vitreous	2 · 2.3	One perfect cleavage; fibrous (satin spar), massive (alabaster), selenite	Scratched by **finger nail**; flexible, not elastic. Plaster of Paris, cement retarder

Quick spotting : soapy = **talc** · nail scratches = **gypsum** · fizzes with HCl = **calcite** · elastic sheets = **mica** · flexible green = **chlorite** · 90° cleavage pink = **orthoclase** · blue blades = **kyanite** · fibres = **asbestos**.

1(b) Ore-Forming Minerals – Identification Answers

Mineral / Formula	Colour · Streak · Lustre	H · SG	Form / Cleavage	Identifying features · Uses
Magnetite Fe_3O_4	Iron-black · **black** · metallic	5.5–6.5 · 5.2	No cleavage; octahedral, massive	**Strongly magnetic**. Best iron ore (72 % Fe)
Hematite Fe_2O_3	Steel-grey to red-brown · **cherry-red / brown** · metallic–earthy	5.5–6.5 · 5.3	No cleavage; kidney (reniform), micaceous, earthy	Red streak. Iron ore (70 % Fe); red ochre pigment
Pyrite FeS_2	Pale brass-yellow ("fool's gold") · greenish-black · metallic	6–6.5 · 5.0	Cubes with **striated faces**; conchoidal, brittle	Strikes sparks with steel. Sulphur / H_2SO_4 source
Pyrolusite MnO_2	Black / dark steel-grey · **black** · metallic–dull	2–2.5 (soft) · 4.7–5	Massive, **dendritic**, powdery; soils fingers	Soils hands black. Manganese ore: steel, dry cells, glass decolouriser
Graphite C	Dark steel-grey/black · black · dull metallic	1–2 · 2.1–2.3	One perfect cleavage; scaly, flaky, earthy	Greasy feel, **marks paper**. Pencil, lubricant, electrodes, crucibles

Mineral / Formula	Colour · Streak · Lustre	H · SG	Form / Cleavage	Identifying features · Uses
Chromite $FeCr_2O_4$	Black / brownish-black · **dark brown** · sub-metallic	5.5 · 4.5–4.8	No cleavage; granular, massive	Weakly magnetic; in ultramafic rocks. Only chromium ore (steel, plating)

Streak is the key : magnetite **black** · hematite **red-brown** · pyrite **greenish-black** · pyrolusite **black (soils fingers)** · graphite **black (marks paper)** · chromite **dark brown.**

Sample Answer – Mineral Description (write-up format)

Property	Specimen : QUARTZ
Colour	Colourless / milky white
Streak	White
Lustre	Vitreous
Hardness	7 (scratches glass)
Cleavage / fracture	Absent / conchoidal
Form	Hexagonal prism with pyramid termination
Specific gravity	2.65
Chemical composition	SiO_2
Uses	Glass, abrasives, concrete aggregate, ornaments

EXPERIMENT 2 – ROCKS

2(a) Igneous Rocks (formed by cooling of magma)

- **Plutonic** (deep, coarse) · **Hypabyssal** (shallow dykes/sills, medium / porphyritic) · **Volcanic** (surface, fine / glassy / vesicular).
- By silica : **Acidic** (> 65 %), **Intermediate** (55–65 %), **Basic** (45–55 %), **Ultrabasic** (< 45 %).

Rock	Texture / Structure	Minerals · Colour	Origin · Formation	Engg. properties & uses
Granite	Coarse, **phaneritic**, holocrystalline, equigranular	Quartz, orthoclase, plagioclase, mica/hornblende · light pink / grey	**Plutonic, acidic**; slow cooling at depth	SG 2.6–2.7; UCS 100–250 MPa; excellent foundation, building / dimension stone, aggregate
Pegmatite	Very coarse (crystals cm–m), irregular intergrowth (graphic)	Quartz, feldspar, mica, tourmaline, beryl	**Dykes / veins** from residual fluid rich in volatiles	Source of mica, feldspar, gems; weak planes along coarse crystals
Gabbro	Coarse, phaneritic, granular	Labradorite + augite ± olivine · dark grey-black	**Plutonic, basic**	SG 2.9–3.1; very hard, tough; road metal, "black granite"
Dolerite	Medium grain, **ophitic** texture (plagioclase laths in augite)	Plagioclase, augite, magnetite · dark	**Hypabyssal basic** (dykes, sills)	Strong, durable; ballast, aggregate, road metal
Syenite	Coarse, phaneritic	Orthoclase dominant + hornblende; **little / no quartz** · pink-grey	Plutonic, **intermediate**	Hard, resists weathering; building and ornamental stone
Granite porphyry	**Porphyritic** : large phenocrysts in fine groundmass	Feldspar / quartz phenocrysts · pink, grey	**Hypabyssal**; two-stage cooling	Strong; dimension stone, aggregate
Basalt	**Aphanitic** fine-grained; vesicular / amygdaloidal; columnar jointing	Plagioclase, augite, olivine · dark grey-black	**Volcanic, basic** (Deccan Traps)	SG 2.8–3.0; UCS 100–300 MPa; road metal, aggregate; dam foundation (watch joints)

2(b) Sedimentary Rocks (deposition + compaction + cementation)

- Features : **stratification (bedding)**, ripple marks, cross-bedding, fossils; layered, porous. Types : clastic (mechanical), chemical, organic.

Rock	Texture / Structure	Minerals · Colour	Origin · Formation	Engg. properties & uses
Sandstone	Clastic, **arenaceous**; bedding, cross-bedding; gritty	Quartz grains + cement (silica / calcareous / ferruginous) · buff, grey, red	Mechanical deposition + cementation	Porous; UCS 20–170 MPa; building / paving stone
Ferruginous sandstone	Same; cement = **iron oxide**	Quartz + limonite / hematite · reddish-brown, purple	Iron-rich cement	Moderately strong; stains water; building stone
Limestone	Non-clastic, fine to crystalline; may have fossils	Calcite ($CaCO_3$) · white, grey, buff	Chemical / organic precipitation	**Effervesces with HCl**; cement, flux, aggregate; cavity / solution hazard
Shale	Argillaceous; **fissile**, laminated; very fine	Clay minerals, silt · grey, black, brown	Compaction of mud / clay	Weak, slakes, swelling; poor foundation / slope stability
Laterite	Porous, vesicular, pitted; soft when fresh, hardens on exposure	Fe & Al hydroxides (goethite, gibbsite) · red-brown	Residual – tropical chemical weathering	Building blocks, road-base; bauxite / iron ore
Conglomerate	**Rudaceous**; rounded pebbles in sandy matrix (angular = breccia)	Pebbles of quartz, quartzite etc. + matrix	Deposition by rivers / beaches	Heterogeneous, variable strength; aggregate

2(c) Metamorphic Rocks (heat + pressure + chemically active fluids)

- Types : **contact (thermal)**, **regional (dynamothermal)**, **dynamic**. Textures : **foliated** (slate → phyllite → schist → gneiss) and **non-foliated** (marble, quartzite).

Rock	Texture / Structure	Minerals · Colour	Origin · Formation	Engg. properties & uses
Biotite-granite gneiss	**Gneissic banding**: alternate light (qtz-feldspar) & dark (biotite) bands; coarse	Quartz, feldspar, biotite · grey-pink	Regional metamorphism of granite	Strong, but weak along foliation; aggregate, building stone

Rock	Texture / Structure	Minerals · Colour	Origin · Formation	Engg. properties & uses
Slate	Very fine; perfect **slaty cleavage**; rings when tapped	Clay-mica, chlorite · grey, black, green	Low-grade regional metamorphism of shale	Splits into slabs: roofing, flooring; poor across cleavage
Muscovite schist	**Schistose** – wavy foliation, silvery flakes; medium-coarse	Muscovite + quartz (± garnet) · silvery	Regional metamorphism of shale / slate	Weak planes → slope failure, poor foundation
Biotite schist	Schistose; dark flaky	Biotite + quartz · dark brown / black	Regional metamorphism	As above; weathers easily
Marble	Granular crystalline **non-foliated**, saccharoidal	Calcite / dolomite · white, pink, green, black	Thermal (contact) metamorphism of limestone	Effervesces with HCl; flooring, sculpture, decorative
Khondalite	Medium-coarse, banded / gneissic	Garnet + sillimanite + graphite + quartz + feldspar · grey-brown	Regional metamorphism of aluminous sediments (Eastern Ghats, Vizag)	Hard, durable; building stone, aggregate

Rock cycle : Igneous → (weathering + deposition) → Sedimentary → (heat/pressure) → Metamorphic → (melting) → Magma → Igneous.

Sample Answer – Rock Description

Item	Specimen : GRANITE
Class	Igneous – plutonic, acidic
Colour	Light pink / grey, speckled
Texture	Coarse-grained, phaneritic, holocrystalline, equigranular
Minerals	Quartz, orthoclase, plagioclase, biotite / hornblende
Structure	Massive, jointed (sheeting)
Engg. use	Foundation rock, building stone, aggregate, dam & bridge piers; UCS 100–250 MPa

EXPERIMENT 3 – GEOLOGICAL MAPS & SECTIONS

3. Interpretation of Geological Maps and Drawing of Sections

Aim : To interpret geological maps (tilted beds, faults, unconformities) and draw the geological section along a given line.

Basic terms

- **Outcrop** – exposure of a bed on the ground; **Contour** – line of equal elevation; **Strike** – direction of horizontal line on a bed; **Dip** – maximum inclination, ⊥ strike, in direction of descent.
- **Dip notation :** 30°→ N60°E means dip of 30° towards N60°E. **Apparent dip** (< true dip) is measured in any other direction.

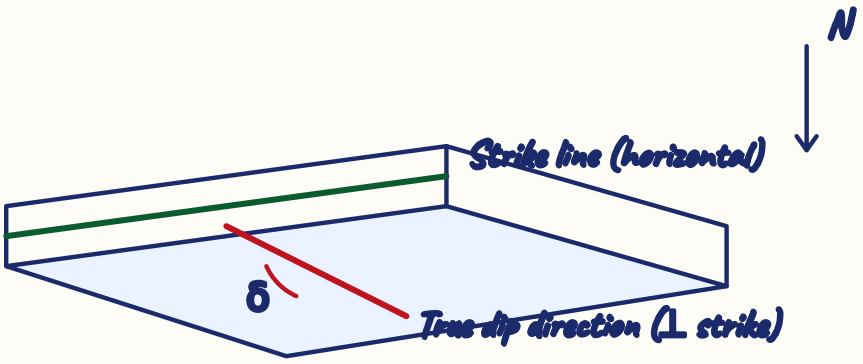


Fig : Strike, dip direction and true dip of a bed

Outcrop patterns – rules (read the map first)

Bed attitude	Outcrop on map
Horizontal	Follows **contour lines** (parallel to them)
Vertical	**Straight line**, unaffected by topography
Dipping	Crosses contours : **V-rule** – in a valley the "V" of outcrop **points in the direction of dip** (for dip steeper than valley slope)
Wide outcrop	Gentle dip / thick bed ; narrow outcrop → steep dip / thin bed
Fold	Repetition of beds in reverse order : **anticline** – older in core; **syncline** – younger in core

Steps – Drawing a geological section along line AB

1. Mark the section line AB on the map; note the points where each contact and contour cuts AB.
2. Draw the **topographic profile** (ground line) : horizontal scale = map scale, vertical elevations from contours (use same vertical scale; mention exaggeration if any).
3. Mark the **contact points** on the profile at their positions along AB.
4. Find dip : strike from outcrop pattern; **true dip** ⊥ strike. If AB is not ⊥ strike use **apparent dip** ($\tan\delta' = \tan\delta \cdot \sin\beta$).
5. From each contact point draw the beds **downward at dip angle** (parallel for conformable beds) ; thicken/continue to sides of the section.
6. Shade / hatch each rock unit, label formations **oldest at bottom**, show dip arrows, faults and unconformities; give title, scale, direction (A→B).

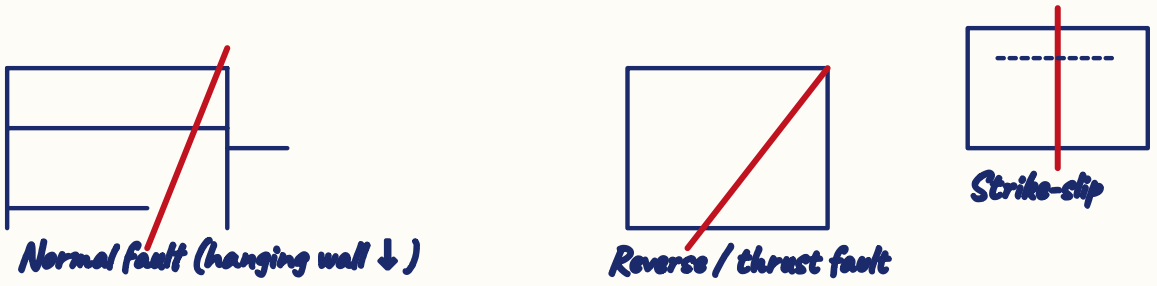


Fig : Types of faults (schematic)

Interpreting structures on a map

- **Fault** – sudden termination / displacement (offset) of beds along a line, repetition or omission of beds. **Normal** : hanging wall moves down (tension); **Reverse/thrust** : hanging wall up (compression); **Strike-slip** : horizontal movement.
- **Unconformity** – gap in rock record; younger beds overlie truncated older beds with a different attitude; **basal conglomerate** at the base. Kinds: angular, disconformity, nonconformity (sediment on igneous/metamorphic).
- **Igneous contacts** – dyke (cuts across beds), sill (parallel to beds), batholith (circular mass).
- **Order of events** (relative age) : beds are older than the faults / dykes that cut them; younger beds lie on top of an unconformity.

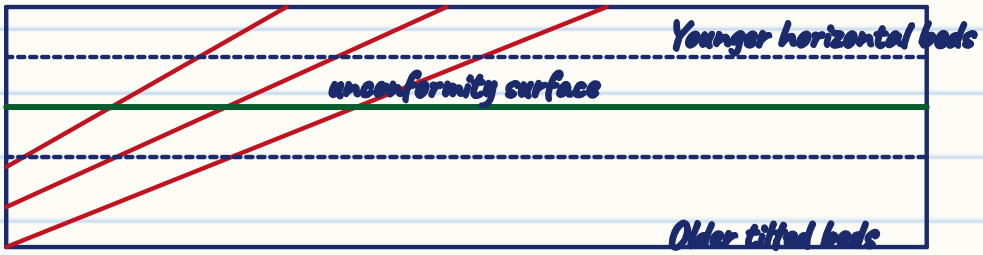


Fig : Angular unconformity

Exam tip : always write "Section along AB – Scale 1 cm = ___ m", show dip symbol and use the legend. Neatness = marks.

EXPERIMENT 4 – STRIKE & DIP PROBLEMS

4. Simple Structural Geology Problems

Apparent dip : $\tan\delta' = \tan\delta \cdot \sin\beta$ (β = angle between strike & apparent-dip direction)

True dip from two apparent dips : use the 3-point method (below) or $\tan\delta = \sqrt{(\tan^2\delta_1 + \tan^2\delta_2)}$ if the two directions are ⊥

Thickness $t = w \cdot \sin\delta$ (horizontal ground) ; vertical thickness = $w \cdot \tan\delta$; depth to bed = $d \cdot \tan\delta$

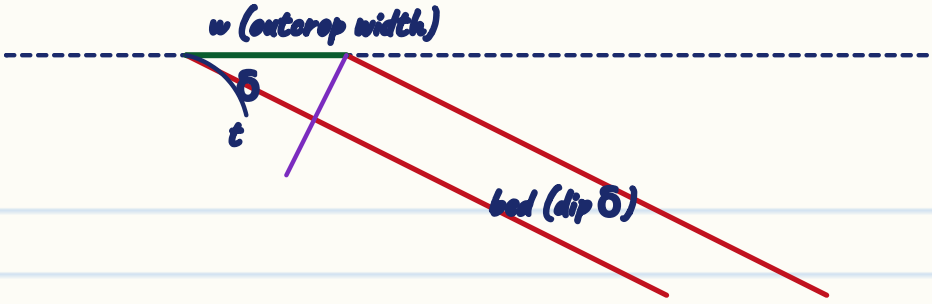


Fig : thickness $t = w \sin\delta$ (horizontal ground) ; vertical thickness = $w \tan\delta$

Problem 1 – Apparent dip

A bed has true dip 30° due east. Find the apparent dip in a direction N45°E (strike is N–S).

- Angle between strike (N–S) and N45°E : $\beta = 45°$
- $\tan\delta' = \tan 30° \times \sin 45° = 0.5774 \times 0.7071 = 0.4082$

$$\delta' = \tan^{-1}(0.4082) = 22.2^\circ \ (< \text{true dip} \ \checkmark)$$

Problem 2 – Thickness from outcrop width

Outcrop width on level ground (measured ⊥ strike) is 100 m ; dip 30°. Find true thickness, vertical thickness.

- True thickness $t = w \sin\delta = 100 \times \sin 30^\circ = 50$ m
- Vertical thickness $= w \tan\delta = 100 \times \tan 30^\circ = 57.7$ m

Problem 3 – Depth of a bed

Bed outcrops at point P; at Q, 200 m away ⊥ strike in the dip direction, dip = 20°. Depth of bed below Q :

$$\text{depth} = 200 \times \tan 20^\circ = 200 \times 0.364 = 72.8 \text{ m}$$

Problem 4 – Three-point problem (strike & dip)

Top of a coal bed is at elevations A = 200 m, B = 100 m, C = 150 m. B is 300 m due east of A; C is 200 m south of AB, 150 m east of A. Find strike, dip & dip direction.

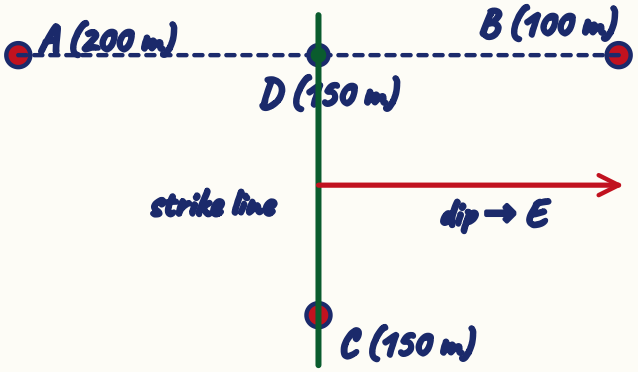


Fig : Three-point (borehole) problem – strike line through D–C

- Point D on AB at elevation 150 m (midway, 150 m east of A). Join D–C (both 150 m) = **strike line** – runs N–S (C is directly south of D).
- Dip is ⊥ strike, towards lower elevation → **due East.**
- Horizontal distance from strike (150 m) to B (100 m) ⊥ strike = 150 m ; drop = 50 m

$$\tan\delta = 50/150 = 0.333 \rightarrow \delta = 18.4^\circ : \text{strike N–S, dip } 18.4^\circ \text{ due E}$$

Procedure (any three-point problem)

1. Plot the 3 points to scale with elevations.
2. Interpolate the point on the line joining highest & lowest that has the elevation of the middle point.
3. Join it to the middle point → **strike line.**

4. Draw ⊥ to strike towards lower side → dip direction; measure horizontal distance d and height difference h : $\tan \delta = h/d$.

EXPERIMENT 5 – BORE-HOLE DATA

5. Bore Hole Problems

- Boreholes give the **depth to each stratum** and **core** – used to find thickness, attitude (strike/dip) of beds, rock quality and structure beneath a site.
- **Vertical borehole through a dipping bed** : measured (vertical) thickness t_v ; true thickness $t = t_v \cos \delta$.
- **Bed elevation** = collar elevation – depth to bed.

Problem 5 – Dip from three boreholes

Three vertical boreholes at corners of a triangle meet the top of a limestone bed at depths : BH1 (collar 300 m) 40 m ; BH2 (collar 280 m) 80 m ; BH3 (collar 290 m) 100 m. Find bed elevations.

- BH1 : 300 – 40 = **260 m** ; BH2 : 280 – 80 = **200 m** ; BH3 : 290 – 100 = **190 m**
- Now treat BH1, BH2, BH3 as points with elevations 260, 200, 190 → use the three-point method to get strike (line through 200 m contour point and 200 m on BH1–BH3… interpolate) and dip.

Rock Quality Designation (RQD) & Core Recovery

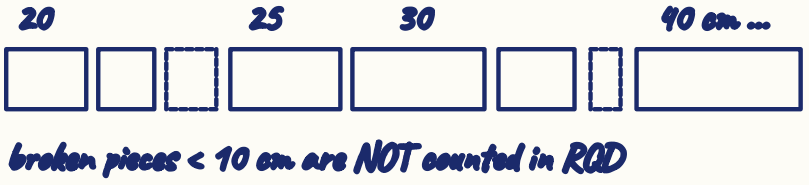


Fig : Core pieces in a run – count only pieces ≥ 10 cm

RQD = Σ (length of core pieces ≥ 10 cm) / total core run × 100 %
Core recovery = length of core recovered / length of run × 100 %

Problem 6 : In a 150 cm run, recovered pieces are 20, 15, 12, 25, 30, 18 cm and some smaller fragments. RQD = (20+15+12+25+30+18)/150 = 120/150 = 80 % → **Good.**

RQD (%)	Rock quality
< 25	Very poor
25–50	Poor
50–75	Fair
75–90	Good

RQD (%)	Rock quality
90–100	Excellent

- **Uses of borehole data :** bedrock depth, weathered zone thickness, water table, cavities, faults/joints, bearing capacity & foundation level (dams, bridges, tunnels).

EXPERIMENT 6 – STRENGTH OF ROCK

6. Strength of Rock using Laboratory Tests

(a) Uniaxial compressive strength (UCS) – IS 9143

- Core / cylinder of L/D = 2–2.5 (NX core 54 mm), ends ground flat; load at 0.5–1.0 MPa/s till failure in compression testing machine.

$$\sigma_c = P / A \text{ (P = failure load, A = cross-section area)}$$

(b) Point load strength index – IS 8764

$$I_s = P / D_e^2 ; D_e^2 = 4A/\pi ; UCS \approx (20–25) \times I_{s(50)}$$

(c) Brazilian (indirect tensile) test

$$\sigma_t = 2P / (\pi D L) \text{ (disc of dia D, thickness L)}$$

(d) Other tests

- **Specific gravity, water absorption, porosity** – IS 1124 (dry wt W_d, saturated wt W_s : absorption = $(W_s - W_d)/W_d \times 100$).
- **Schmidt (rebound) hammer** – quick field estimate of strength.
- **Aggregate tests** – Los Angeles abrasion, impact, crushing value (road metal).
- **Slake durability** – shale / weathered rocks.

Problem 7

A 54 mm dia core fails at 280 kN. UCS = $280 \times 10^3 / (\pi/4 \times 54^2) = 280000 / 2290 =$ 122 MPa → **very strong** rock.

UCS (MPa)	Strength class (IS 12070)	Typical rock
< 1	Extremely weak	Weathered / soft
1–5	Very weak	Chalk, weathered shale
5–25	Weak	Shale, soft sandstone
25–50	Medium strong	Limestone, sandstone
50–100	Strong	Marble, gneiss
100–250	Very strong	Granite, basalt, quartzite
> 250	Extremely strong	Fresh dolerite, gabbro

EXPERIMENT 7 – FIELD WORK

7. Field Work – Minerals, Rocks, Geomorphology & Structures

Aim : To identify minerals, rocks, landforms and geological structures in the field and prepare a report.

Equipment : Brunton / clinometer compass, hammer, hand lens, dil. HCl, topo-sheet, GPS, notebook, camera, sample bags, measuring tape.

Observations to record at every station

- **Location** (lat-long), elevation, date; **rock type** (igneous / sedimentary / metamorphic), colour, texture, minerals, degree of weathering.
- **Structures :** bedding / foliation **strike & dip** (e.g. N30°E / 25° SE), joint sets (spacing, aperture, persistence), folds, faults, dykes, contacts.
- **Geomorphology :** drainage pattern (dendritic, trellis, radial), terraces, alluvial fans, floodplain, deltas, hills, escarpments, sinkholes.
- **Topography :** contours, **slope** (steepness), **aspect** (direction a slope faces) and relief – used to select the site and construction materials.

Measuring strike & dip with a compass-clinometer

1. Place the compass edge on the bed surface, level the bubble → read azimuth = **strike.**
2. Place clinometer ⊥ strike on the bed and read maximum angle = **dip;** note the dip direction (e.g. N60°E).

Importance in civil engineering

- **Dams** – avoid faults, shear zones, cavernous limestone, dips towards downstream/valley side.
- **Tunnels** – rock mass quality (RQD), joints, groundwater; **Roads / slopes** – bedding dipping out of the cut → landslides.
- **Foundations** – sound, unweathered rock (granite, basalt, gneiss) preferred; shale, schist, cavernous limestone, laterite are poor.

Format of the project report on geology

1. Title page · Certificate · Acknowledgement · Abstract
2. Introduction – location, objective, accessibility, climate

3. Geology of the area – physiography, drainage, rocks (with mineral descriptions), structures
4. Data tables – strike/dip readings, sections, geological map, photographs
5. Engineering significance & conclusions · References

Viva Voce – Short Q&A

Question	Answer
Hardest and softest minerals?	Diamond (10) and talc (1)
Which mineral has hardness varying with direction?	Kyanite (5 ∥ length, 7 ⊥ length)
Which mica is black?	Biotite; muscovite is colourless / silvery
Mineral that fizzes with dil. HCl?	Calcite (also limestone, marble)
Magnetic ore of iron?	Magnetite
Rock with ophitic texture?	Dolerite
Fine-grained volcanic basic rock?	Basalt (Deccan traps)
Metamorphic rock from limestone / shale / granite?	Marble / slate / gneiss
Rock showing bedding & fossils?	Sedimentary (limestone, shale)
Rock giving rise to laterite soils?	Weathering of basalt / khondalite in tropical climate
Dip vs apparent dip?	True dip is maximum (⊥ strike); apparent dip is less
Why RQD?	Measures quality / fracturing of rock mass from core
Best foundation rocks?	Granite, basalt, gneiss, quartzite, massive limestone (uncavernous)

Best of luck for your lab exam! Revise the identifying features table and practise one three-point and one section-drawing problem.